

DATA-DRIVEN WATER QUALITY SURVEILLANCE

SENSORS SPECIFICATION DOCUMENT

1. Sensor specification

This document contains the sensor specifications for the Water Quality Monitoring System. It is prepared by the documentation team to maintain clarity and consistency across the project.

The following table includes an overview of all sensors and key hardware components used in the project along with their purpose and output.

Overview of sensors and hardware components. :

ID	COMPONENT NAME	TYPE	CATEGORY	PURPOSE	RANGE	UNIT	OUTPUT
S1	Flowmeter Sensor	Digital	Sensor	Measure water flow rate	1-30	L/min	Pulse
S2	TDS Sensor	Analog	Sensor	Measure dissolved solids	0-1000	ppm	Analog
S3	Turbidity Sensor	Analog	Sensor	Measure water clarity	0-1000	NTU	Analog
S4	Ultrasonic Sensor	Digital	Sensor	Measure water level	2-400	cm	Digital
H1	ESP32	Digital	MCU	Process data & control system	–	–	Digital

ESP32 (Microcontroller Unit)

Detailed specification of component :

Field	Details
Component Name	ESP32 Development Board
Component ID	H1
Type	MCU (Microcontroller Unit)
Location	Main control unit of the system
Purpose	Collects sensor data and sends it to the cloud/database via Wi-Fi; acts as the brain of the system Processor Dual-core CPU
Processor	Dual-core CPU
Clock Speed	Up to 240 MHz
Wi-Fi	802.11 b/g/n
Bluetooth	BLE + Classic Bluetooth
Operating Voltage	3.3V
Flash Memory	4MB
Temperature Range	-40°C to 85°C
Humidity Range	5%–95% RH
Output Type	Digital
Data Format	Integer / Float
Sampling Rate	Depends on connected sensors
Data Usage	Real-time monitoring, data analysis, alert generation
Compatible Sensors	TDS, Turbidity, Flowmeter
Pins Required	GPIO pins
Power Requirement	5V USB supply
Connection Type	Wi-Fi / UART / Analog / Digital

FLOWMETER SENSOR (WATER FLOW SENSOR)

Detailed specification of component :

Field	Details
Component Name	Water Flowmeter Sensor
Component ID	S1
Type	Digital Sensor
Location	Water pipeline / Inlet Pipe
Purpose	Measures the flow rate of water in the system
Working Principal	Uses Hall Effect to detect water flow speed
Measurement Range	1 - 30 L/min
Unit	Liters per minute (L/min)
Output Type	Pulse Digital Output
Operating Voltage	5V
Current Consumption	Approx. 15mA
Accuracy	±10%
Response Time	Real-Time
Pipe Compatibility	½ inch pipe
Temperature Range	Up to 80°C
Connection Type	Digital GPIO
Compatible MCU	ESP32 / Arduino
Data Usage	Water usage monitoring and leakage detection
Sampling Rate	Continuous Monitoring
Pins Required	VCC, GND, Signal
Power Requirement	5V DC supply

TURBIDITY SENSOR

Detailed specification of component :

Field	Details
Component Name	Turbidity Sensor
Component ID	S3
Type	Analog Sensor
Location	Water storage / Pipeline
Purpose	Measures water clarity and dirt level
Working Principal	Uses light transmission and scattering
Measurement Range	0 - 1000 NTU
Unit	NTU
Output Type	Analog
Operating Voltage	5V
Accuracy	High Sensitivity
Response Time	Real-Time
Detection Method	Optical Sensing
Data Format	Analog voltage
Compatible MCU	ESP32 / Arduino
Connection Type	Analog GPIO
Data Usage	Detects muddy or contaminated water
Sampling Rate	Continuous
Temperature Range	50°C to 90°C
Pins Required	VCC, GND, Signal
Power Requirement	5V DC supply

ULTRASONIC SENSOR

Detailed specification of component :

Field	Details
Component Name	Ultrasonic Sensor HC-SR04
Component ID	S4
Type	Digital Sensor
Location	Top of the water tank
Purpose	Measures water level and distance
Working Principal	Uses ultrasonic sound waves
Measurement Range	2 - 400 cm
Unit	cm
Output Type	Digital
Operating Voltage	5V
Accuracy	± 3mm
Frequency	40kHz
Response Time	Fast
Detection Type	Non - contact
Compatible MCU	ESP32 / Arduino
Connection Type	Trigger and Echo Pins
Data Usage	Tank level monitoring and overflow detection
Sampling Rate	Real - Time
Pins Required	VCC, GND, Trig. , Echo
Power Requirement	5V DC supply

TDS SENSOR (TOTAL DISSOLVED SOLID SENSORS)

Detailed specification of component :

Field	Details
Component Name	TDS Sensor Module
Component ID	S2
Type	Analog Sensor
Location	Water Tank / Monitoring Chamber
Purpose	Measures dissolved solids in water
Working Principal	Measures electrical conductivity of water
Measurement Range	0 - 1000 ppm
Unit	ppm
Output Type	Analog
Operating Voltage	3.3V - 5V
Accuracy	±10%
Response Time	Fast Response
Temperature Compensation	Supported
Probe Type	Waterproof Probe
Data Format	Integer / Float
Compatible MCU	ESP32 / Arduino
Connection Type	Analog Pin
Data Usage	Water Purity Analysis
Sampling Rate	Depends on Monitoring frequency
Pins Required	VCC, GND, AOUT
Power Requirement	USB / DC supply

CONVERTER MODULE 230V AC - 5V DC

Detailed specification of component :

Field	Details
Component Name	CONVERTER MODULE 230V AC - 5V DC
Component ID	P1
Type	AC to DC Converter Module
Input Voltage	230V AC , 50/60Hz
Input Voltage Range	100 - 265V AC
Output Voltage	5V DC
Output Current	2A (Max)
Output Power	10W (Max)
Output Type	Regulated DC Output
Efficiency	$\geq 80\%$
Ripple and Noise	$\leq 50\text{mV}$
Operating Temperature	- 10°C to 60°C
Storage Temperature	- 20°C to 85°C
Protection	Overall Protection , Short Circuit Protection, Over Voltage Protection
Isolation	High Voltage Isolation
Material	ABS Plastic / PCB
Indicators	Power LED Indicator
Connection Type (Input)	2 Pin Screw Terminal (L,N)
Connection Type (Output)	2 Pin Screw Terminal (+5V, GND)
Mounting Type	PCB Mount / Enclosed Module
Applications	Power supply for microcontrollers, sensors and Low power electronic circuits
Compatible Devices	ESP32, Arduino, Sensors, Modules

Dimensions	50mm x 30mm x 20mm
Weight (Approx.)	30g
Power Requirement	230V AC Main Supply
Certifications	CE, RoHS